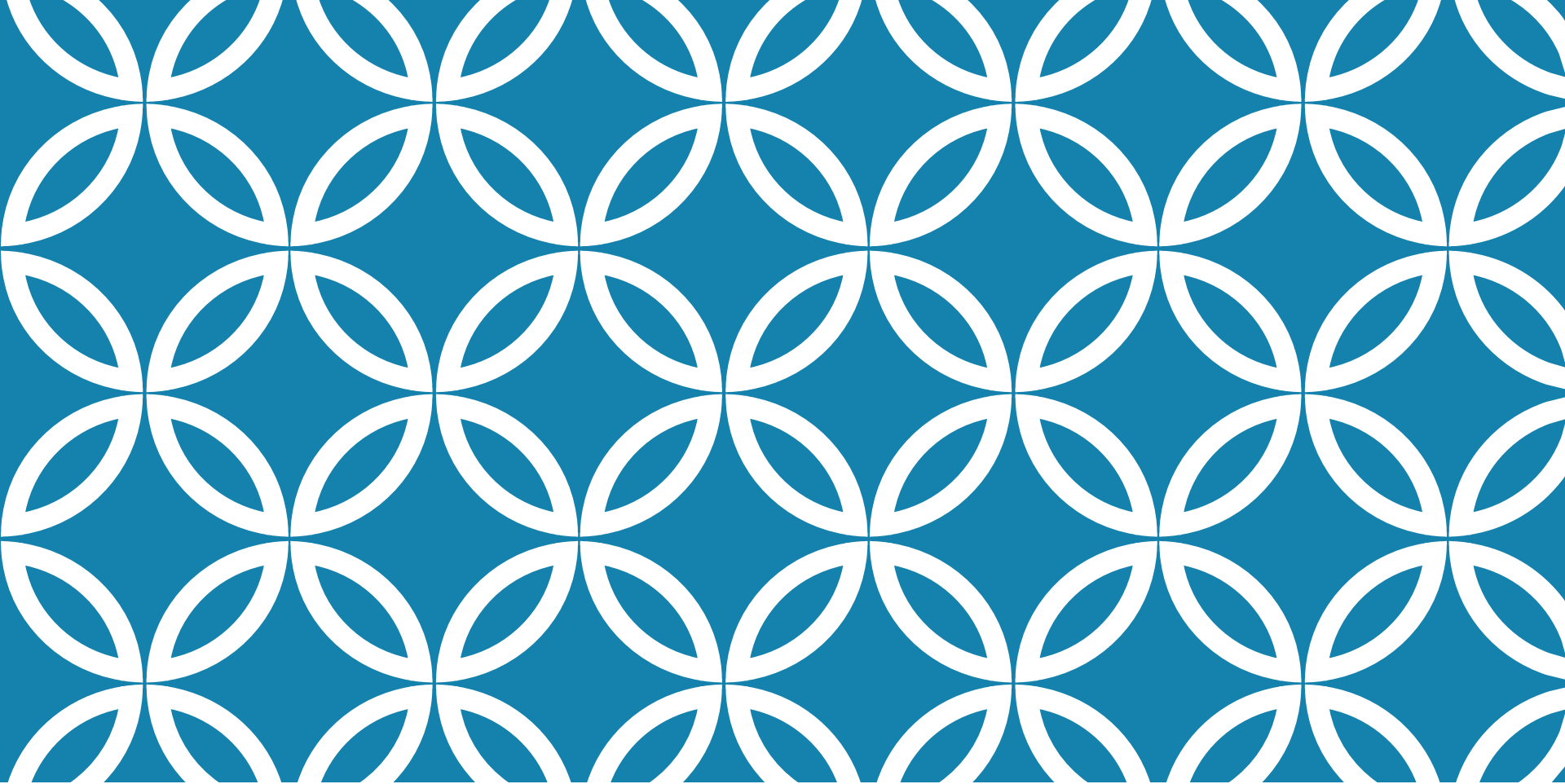




Business Intelligence: Data Warehousing

Dr Tamal Ghosh
Associate Professor
Department of CSE
Adamas University

Lecture Note 2

MAIN DATA WAREHOUSING TOPICS

DW definition

Characteristics of DW

Data Marts

ODS, EDW, Metadata

DW Framework

DW Architecture & ETL Process

DW Development

DW Issues

WHAT IS A DATA WAREHOUSE?

A physical repository where relational data are specially organized to provide enterprise-wide, cleansed data in a standardized format

“The data warehouse is a collection of integrated, subject-oriented databases designed to support DSS functions, where each unit of data is non-volatile and relevant to some moment in time”

CHARACTERISTICS OF DW

Subject oriented

Integrated

Time-variant (time series)

Nonvolatile

Summarized

Metadata

Web based, relational/multi-dimensional

Client/server

Real-time and/or right-time (active)

DATA MART

A departmental data warehouse that stores only relevant data

- **Dependent data mart**

A subset that is created directly from a data warehouse

- **Independent data mart**

A small data warehouse designed for a strategic business unit or a department

DATA WAREHOUSING DEFINITIONS

Operational data stores (ODS)

A type of database often used as an interim area for a data warehouse

Operational marts

An operational data mart

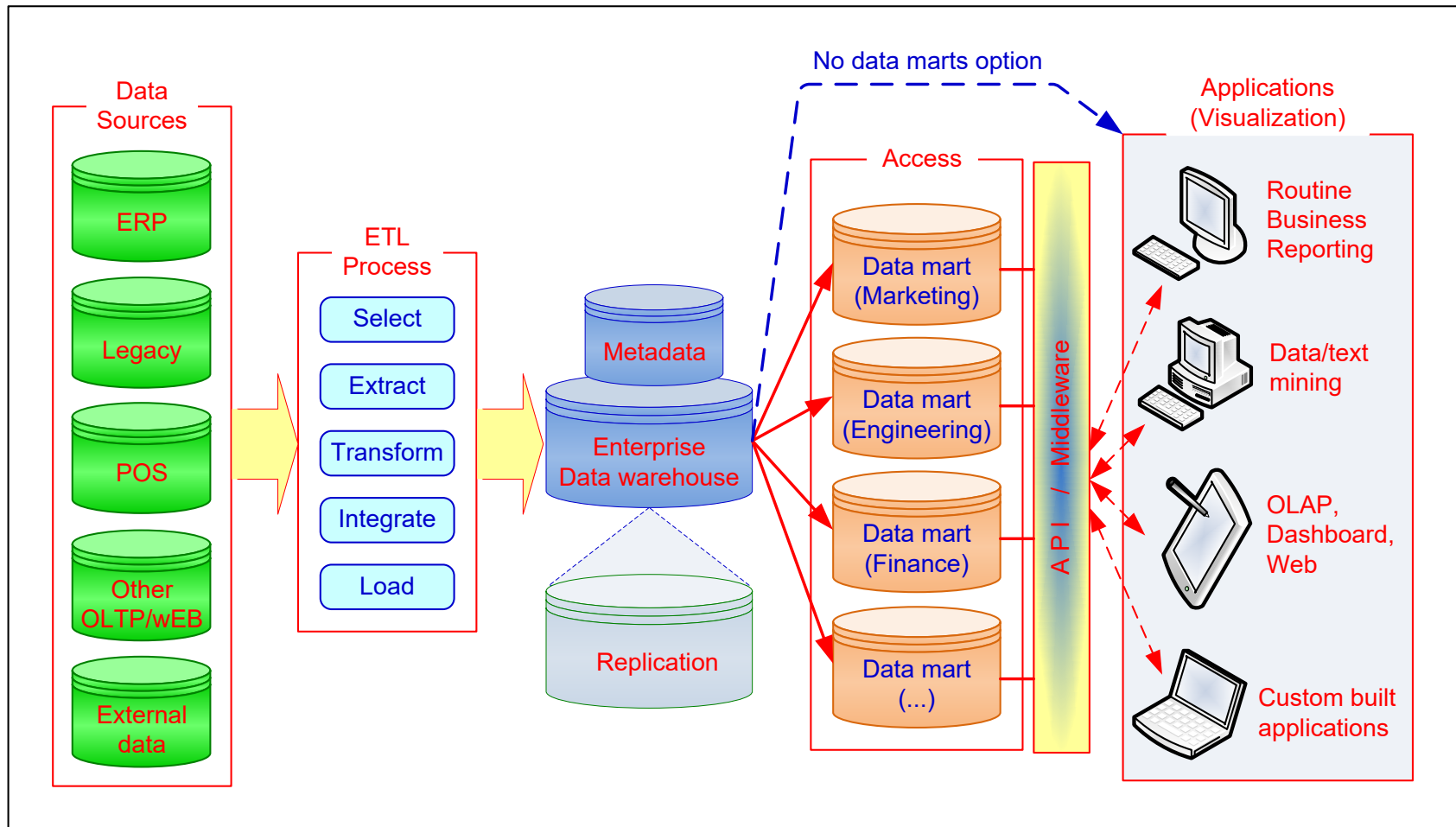
Enterprise data warehouse (EDW)

A data warehouse for the enterprise

Metadata

Data about data. In a data warehouse, metadata describe the contents of a data warehouse and the manner of its acquisition and use

DW FRAMEWORK



DW ARCHITECTURE

Three-tier architecture

1. Data acquisition software (back-end)
2. The data warehouse that contains the data & software
3. Client (front-end) software that allows users to access and analyze data from the warehouse

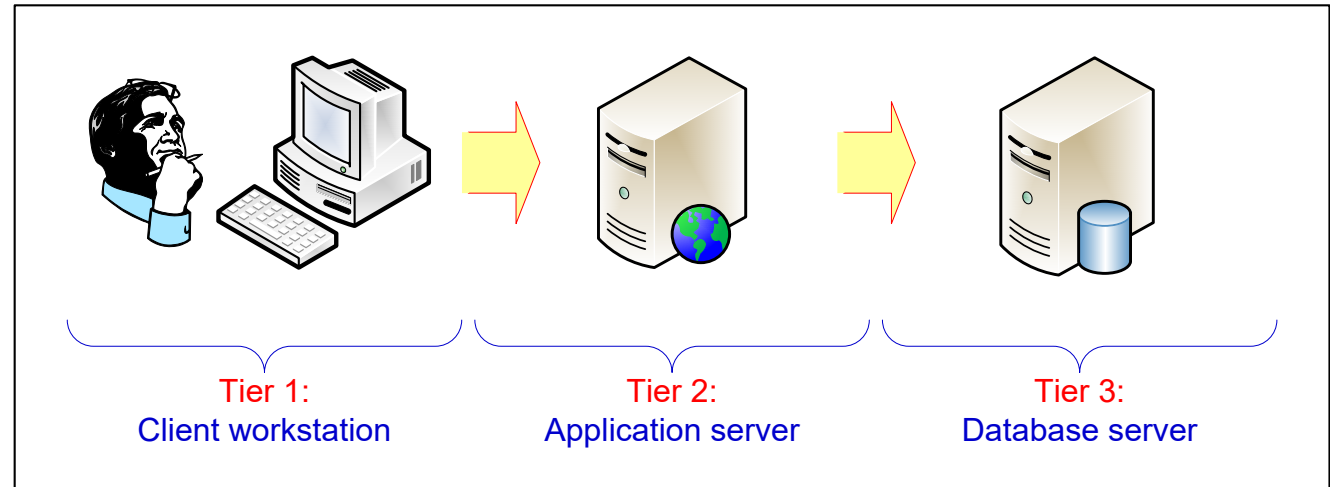
Two-tier architecture

First 2 tiers in three-tier architecture is combined into one

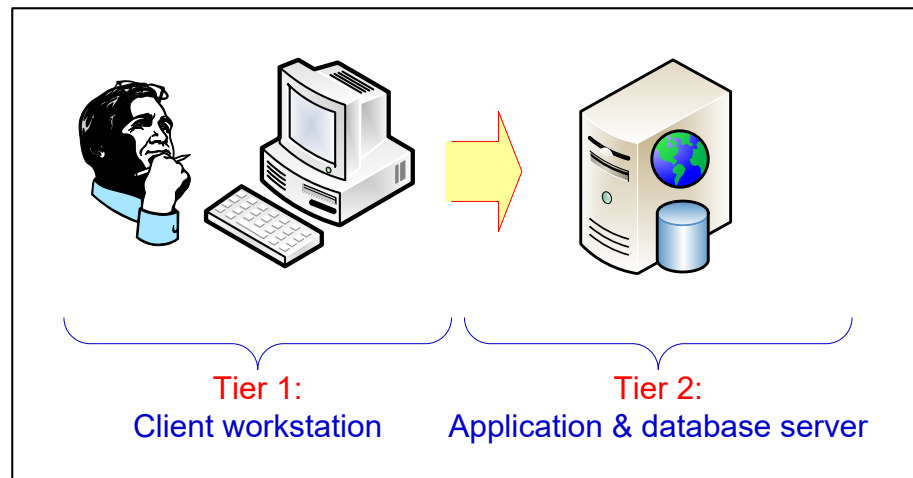
Sometimes there is only one tier

DW ARCHITECTURES

3-tier architecture

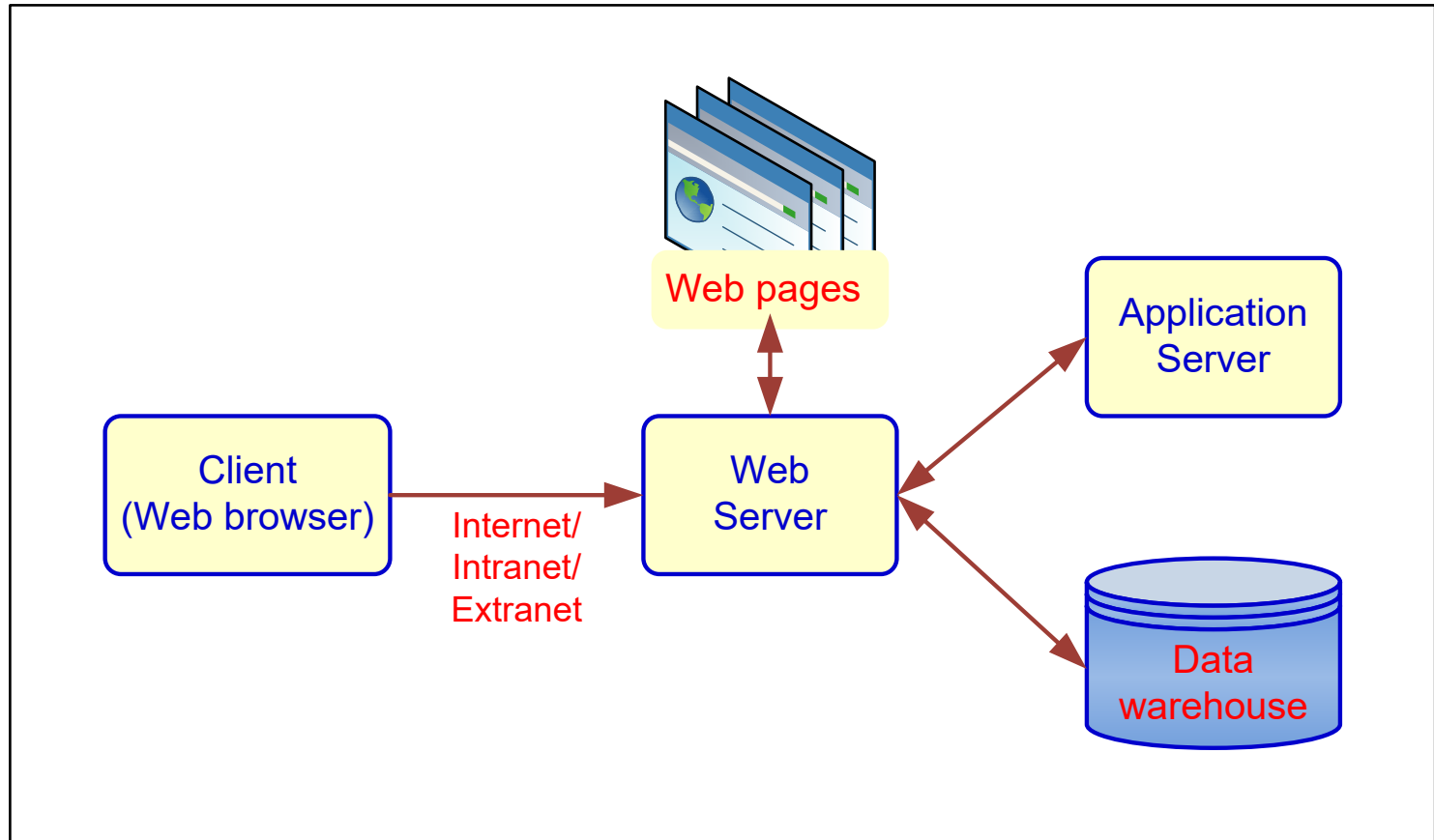


2-tier architecture



1-tier Architecture

A WEB-BASED DW ARCHITECTURE



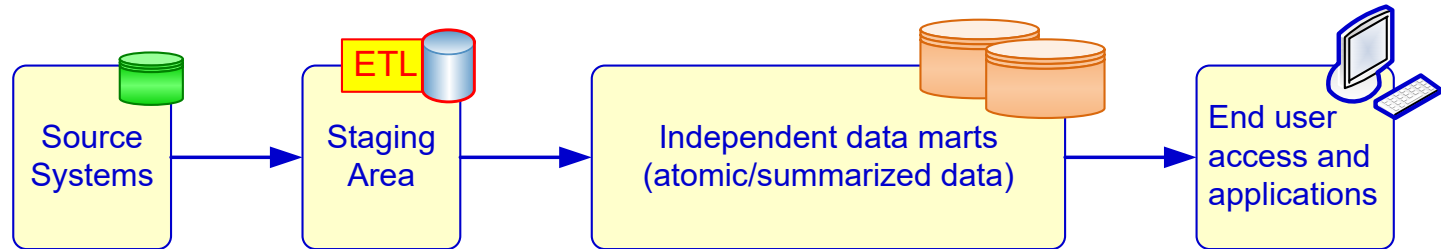
DATA WAREHOUSING ARCHITECTURES

Issues to consider when deciding which architecture to use:

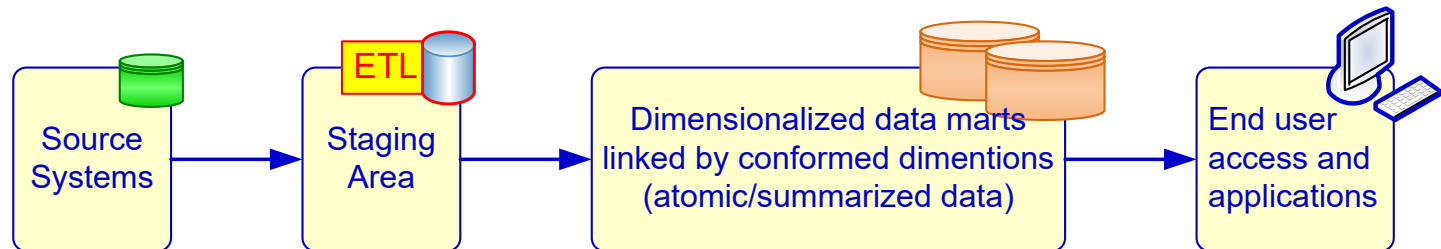
- Which database management system (DBMS) should be used?
- Will parallel processing and/or partitioning be used?
- Will data migration tools be used to load the data warehouse?
- What tools will be used to support data retrieval and analysis?

ALTERNATIVE DW ARCHITECTURES

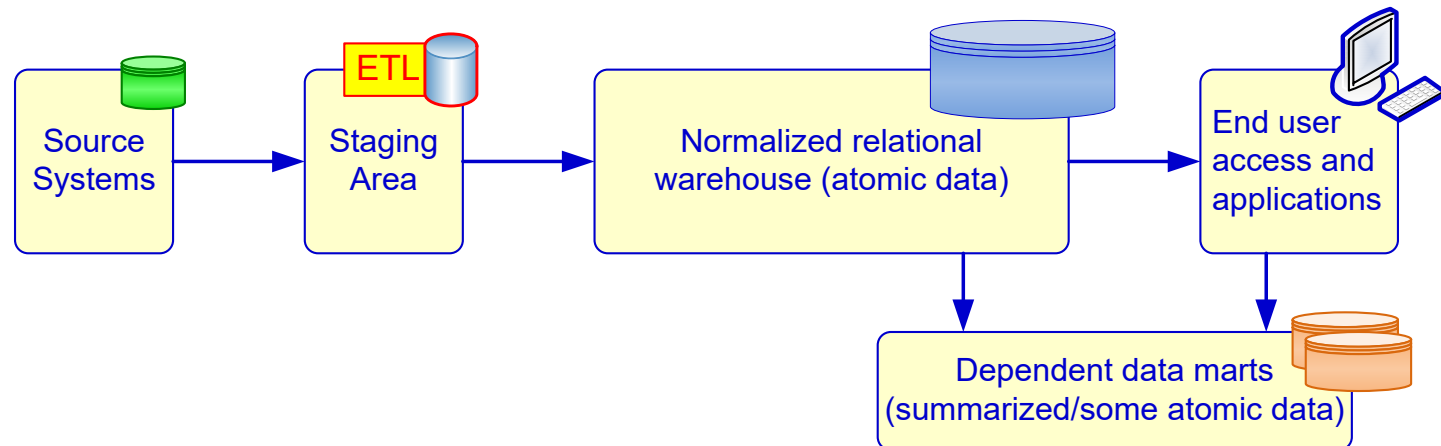
(a) Independent Data Marts Architecture



(b) Data Mart Bus Architecture with Linked Dimensional Datamarts

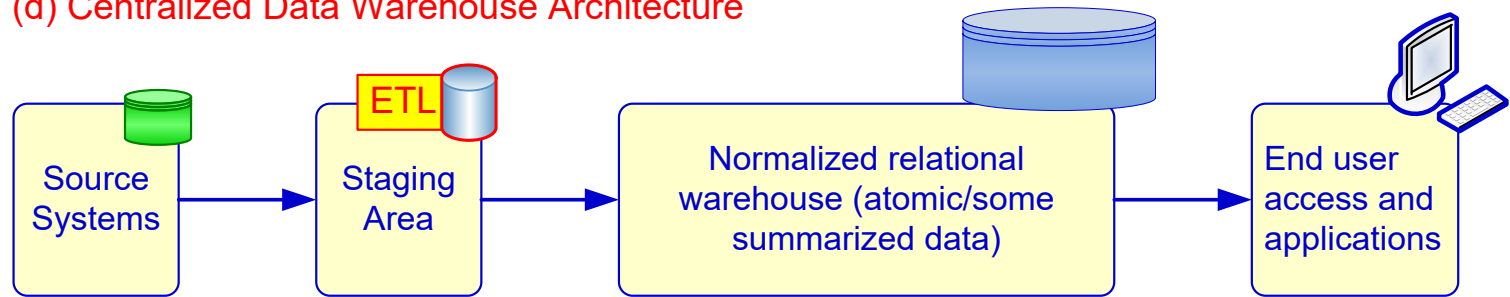


(c) Hub and Spoke Architecture (Corporate Information Factory)

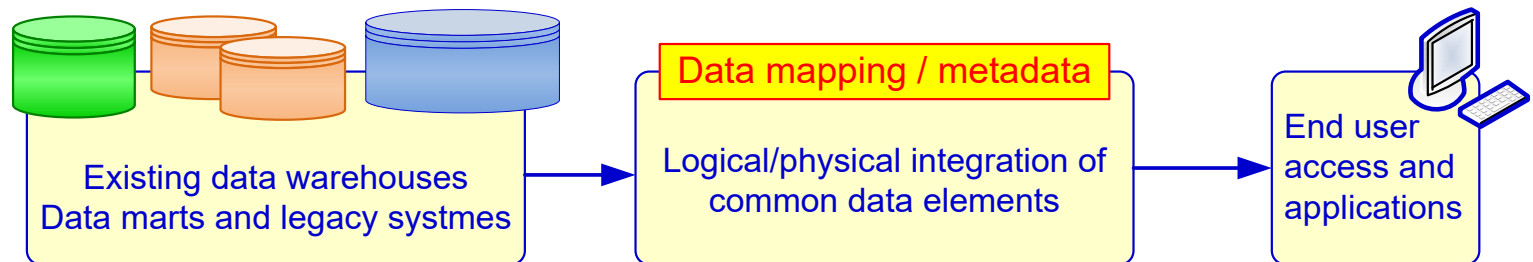


ALTERNATIVE DW ARCHITECTURES

(d) Centralized Data Warehouse Architecture



(e) Federated Architecture

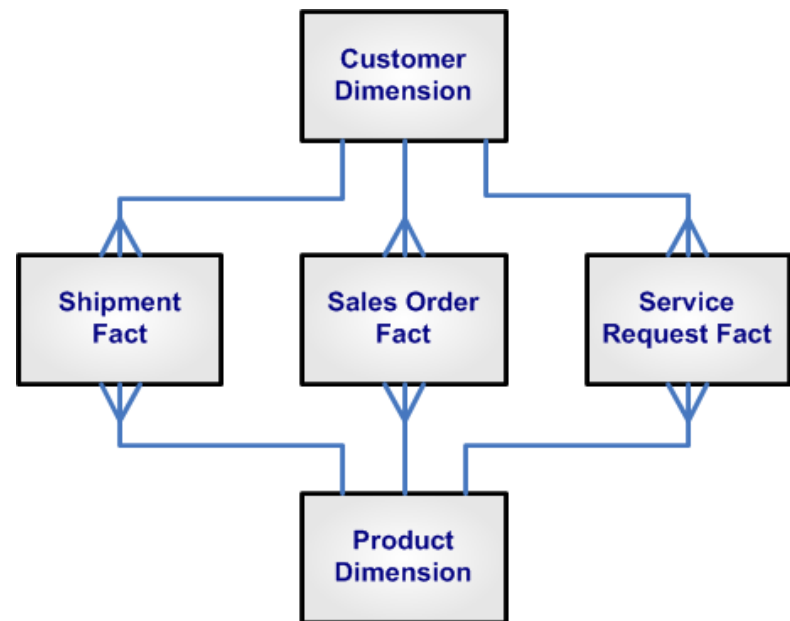


CONFORMED DIMENSIONS

dimensions that have the same meaning and content across multiple data marts and data sources in an organization.

Conformed dimensions allow facts and measures to be categorized and described in the same way across multiple facts or data marts, ensuring consistent reporting across the enterprise.

For example, a conformed dimension of time can have the same hierarchy of year, quarter, month, day, and hour for sales, inventory, and budget facts. This enables you to join and compare data from different business processes or perspectives.



ALTERNATIVE DW ARCHITECTURES

1. Independent Data Marts
2. Data Mart Bus Architecture
3. Hub-and-Spoke Architecture
4. Centralized Data Warehouse
5. Federated Data Warehouse

Each has pros and cons!

TERADATA

Teradata is an open-source DBMS for developing large-scale data warehousing applications.

This tool provides support for multiple DW operations simultaneously using the concept of parallelism.

Teradata is a massively open processing system that supports Unix/Linux/Windows server platforms.

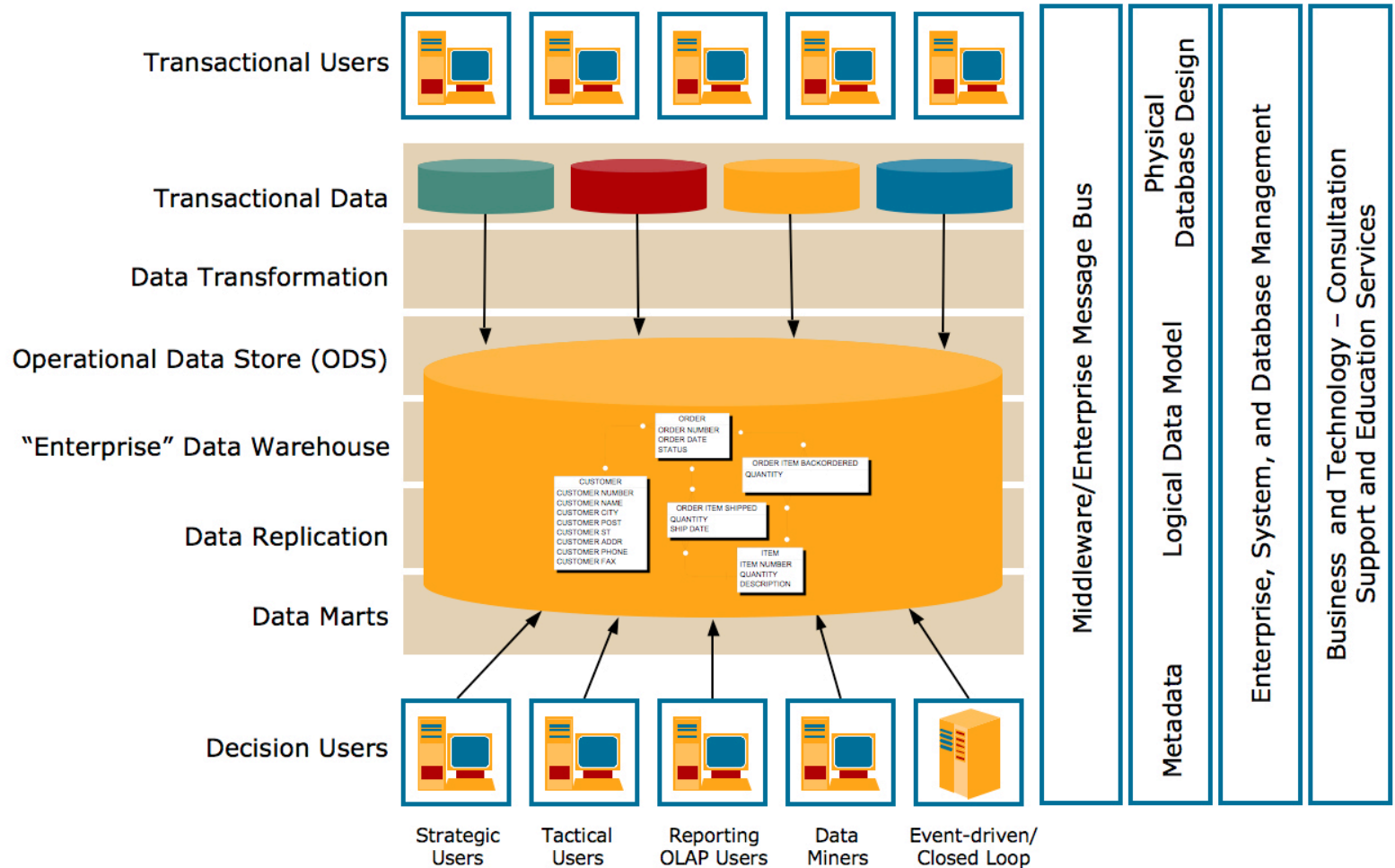
Teradata software is developed by Teradata Corporation, which is an American IT firm.

The firm develops a product to consolidate data from various sources and make the data available for analysis.

TERADATA

- Teradata offers a full suite of service which focuses on [Data Warehousing](#)
- The system is built on open architecture. So whenever any faster devices are made available, it can be incorporated into the already build architecture.
- Teradata supports 50+ petabytes of data.
- Single operation view for a large Teradata multi-node system using Service Workstation
- Compatible with wide range of [BI tool](#) to fetch data.
- It can act as a single point of control for the DBA to manage the [Database](#).
- High performance, diverse queries, in-database analytics and sophisticated workload management
- Teradata allows you to get the same data on multiple deployment options

TERADATA DW ARCHITECTURE



DIFFERENCE BETWEEN TERADATA AND OTHER RDBMS

Parameter	TERA DATA	RDBMS
Architectures	Follows Shared Nothing Architecture.	Shared Everything and allows resource contention.
Processes	MIPS [Millions of Instructions/sec]	KIPS [Thousands of institutions/sec]
Indexes	Better Distribution and Retrieval	Only offers FASI Retrieval
Parallelism	Supports Un-conditional parallelism.	Parallelism is conditional and unpredictable
Bulk Load	Teradata allows bulk load.	Allows only limited bulk load.
Scalability	Linear scalability with a slope of one	Scalability with diminishing returns
Database buffer	A single database buffer used by all UoP's. (A unite of parallelism). A single data store accessed by all UoP's	Query Controller ships functions to UoP's that own the data
Stores	It stores TERA BYTES[Billions of rows]	GIGA BYTES[Millions of rows]

DATA WAREHOUSING ARCHITECTURES

Ten factors that potentially affect the architecture selection decision:

1. Information interdependence between organizational units
2. Upper management's information needs
3. Urgency of need for a data warehouse
4. Nature of end-user tasks
5. Constraints on resources
6. Strategic view of the data warehouse prior to implementation
7. Compatibility with existing systems
8. Perceived ability of the in-house IT staff
9. Technical issues
10. Social/political factors

DATA INTEGRATION AND THE EXTRACTION, TRANSFORMATION, AND LOAD (ETL) PROCESS

Data integration

Integration that comprises three major processes: data access, data federation, and change capture

Enterprise application integration (EAI)

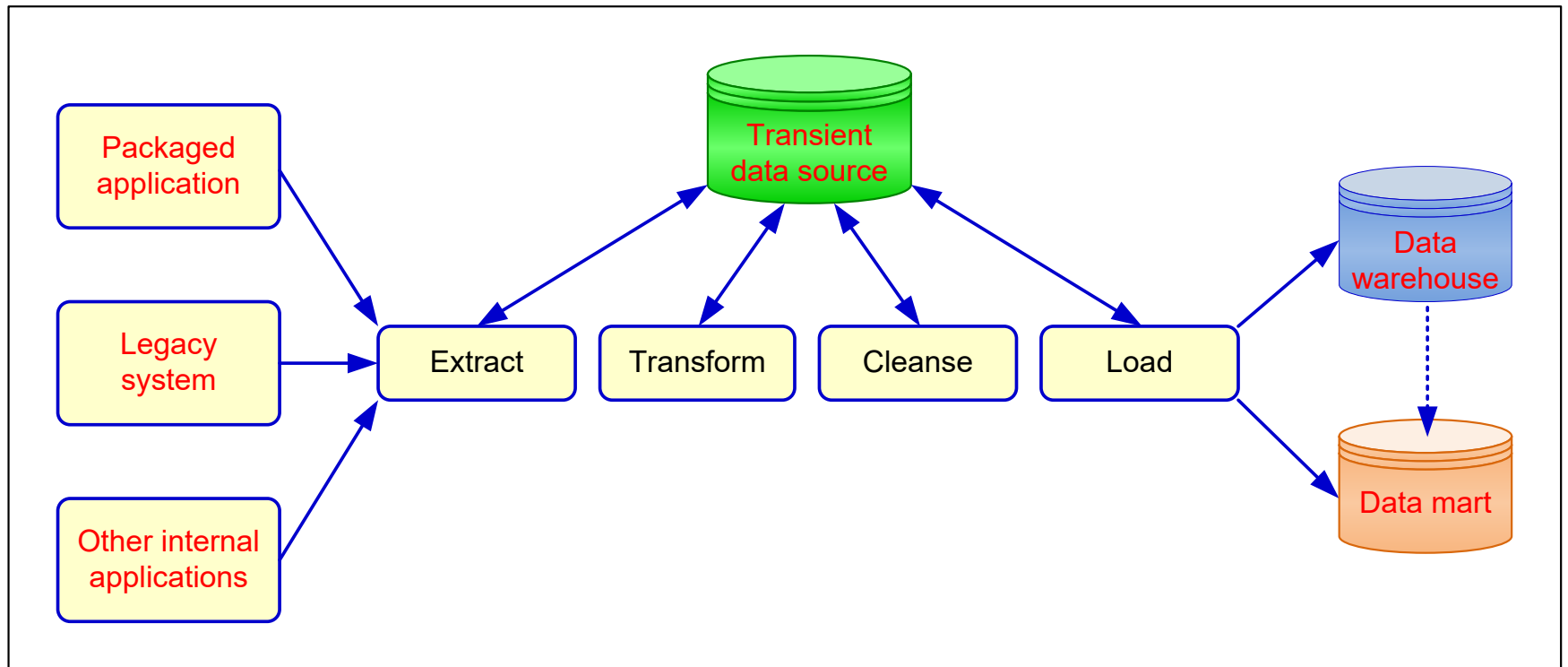
A technology that provides a vehicle for pushing data from source systems into a data warehouse

Enterprise information integration (EII)

An evolving tool space that promises real-time data integration from a variety of sources, such as relational databases, Web services, and multidimensional databases

DATA INTEGRATION AND THE EXTRACTION, TRANSFORMATION, AND LOAD (ETL) PROCESS

Extraction, transformation, and load (ETL)



ETL

Issues affecting the purchase of ETL tool

- Data transformation tools are expensive
- Data transformation tools may have a long learning curve

Important criteria in selecting an ETL tool

- Ability to read from and write to an unlimited number of data sources/architectures
- Automatic capturing and delivery of metadata
- A history of conforming to open standards
- An easy-to-use interface for the developer and the functional user

DATA WAREHOUSE DEVELOPMENT

Data warehouse development approaches

- Inmon Model: EDW approach (top-down)
- Kimball Model: Data mart approach (bottom-up)
- Which model is best?
 - There is no one-size-fits-all strategy to DW
- One alternative is the hosted warehouse

Data warehouse structure:

- The Star Schema vs. Relational

Real-time data warehousing?

HOSTED DATA WAREHOUSES

Benefits:

- Requires minimal investment in infrastructure
- Frees up capacity on in-house systems
- Frees up cash flow
- Makes powerful solutions affordable
- Enables powerful solutions that provide for growth
- Offers better quality equipment and software
- Provides faster connections
- Enables users to access data remotely
- Allows a company to focus on core business
- Meets storage needs for large volumes of data

REPRESENTATION OF DATA IN DW

Dimensional Modeling – a retrieval-based system that supports high-volume query access

Star schema – the most commonly used and the simplest style of dimensional modeling

- Contain a **fact table** surrounded by and connected to several **dimension tables**
- Fact table contains the descriptive attributes (numerical values) needed to perform decision analysis and query reporting
- Dimension tables contain classification and aggregation information about the values in the fact table

Snowflakes schema – an extension of star schema where the diagram resembles a snowflake in shape

MULTIDIMENSIONALITY

Multidimensionality

The ability to organize, present, and analyze data by several dimensions, such as sales by region, by product, by salesperson, and by time (four dimensions)

Multidimensional presentation

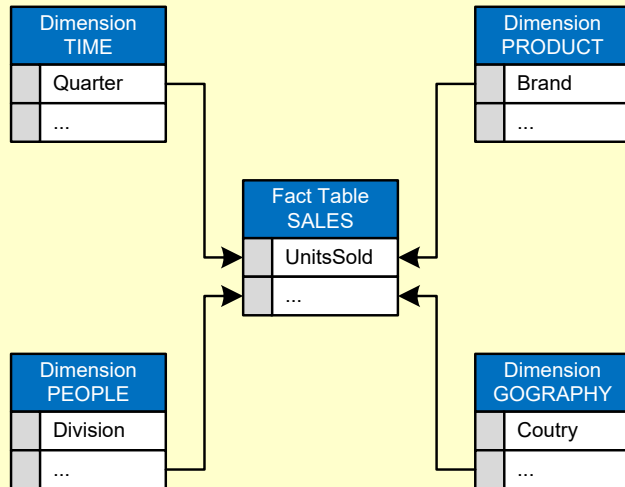
- **Dimensions:** products, salespeople, market segments, business units, geographical locations, distribution channels, country, or industry
- **Measures:** money, sales volume, head count, inventory profit, actual versus forecast
- **Time:** daily, weekly, monthly, quarterly, or yearly

STAR VS SNOWFLAKE SCHEMA

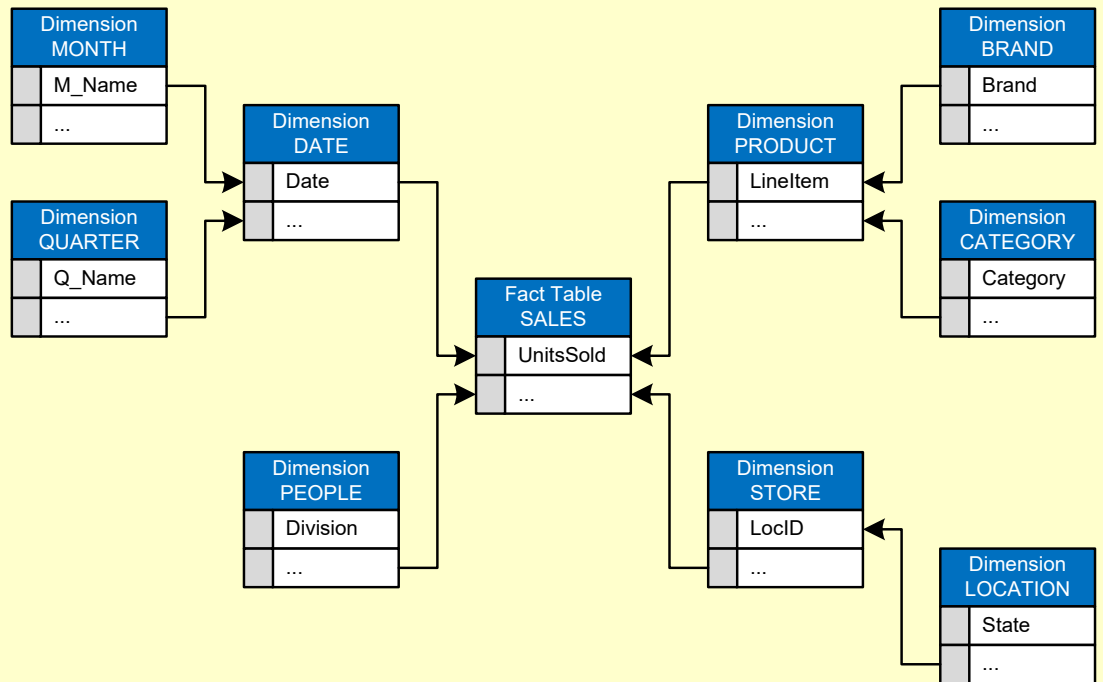
Feature	Star Schema	Snowflake Schema
Structure	Central fact table connected directly to dimension tables.	Central fact table connected to normalized dimension tables (with sub-dimensions).
Normalization Level	Denormalized (flat structure).	Partially normalized (split into multiple related tables).
Complexity	Simple and easy to design and query.	More complex due to multiple related tables.
Query Performance	Faster queries due to fewer joins.	Slower queries due to multiple joins between tables.
Storage Space	Requires more storage due to data redundancy.	Requires less storage as it reduces data redundancy.
Maintenance	Easier to maintain and update.	More difficult to maintain and update.
Use Case	Suitable for small to medium-sized databases and simpler queries.	Suitable for large databases with complex queries.
Dimension Tables	Single-level dimension tables.	Hierarchical structure with multiple levels in dimension tables.
Design Complexity	Easier design with fewer tables.	More complex design with more tables and relationships.
Data Redundancy	High (redundant data stored in dimension tables).	Low (redundancy is minimized through normalization).
Example	Sales fact table connected to product, time, and customer dimensions.	Sales fact table with normalized product dimension split into product, category, and brand tables.

STAR VS SNOWFLAKE SCHEMA

Star Schema



Snowflake Schema



ANALYSIS OF DATA IN DW

Online analytical processing (OLAP)

- Data driven activities performed by end users to query the online system and to conduct analyses
- Data cubes, drill-down / rollup, slice & dice, ...

OLAP Activities

- Generating queries (query tools)
- Requesting ad hoc reports
- Conducting statistical and other analyses
- Developing multimedia-based applications

ANALYSIS OF DATA STORED IN DW

OLTP VS. OLAP

OLTP (online transaction processing)

- A system that is primarily responsible for capturing and storing data related to day-to-day business functions such as ERP, CRM, SCM, POS,
- The main focus is on efficiency of routine tasks

OLAP (online analytic processing)

- A system is designed to address the need of information extraction by providing effectively and efficiently ad hoc analysis of organizational data
- The main focus is on effectiveness

OLAP VS. OLTP

Criteria	OLTP	OLAP
Purpose	To carry out day-to-day business functions.	To support decision making and provide answers to business and management queries.
Data source	Transaction database (a normalized data repository primarily focused on efficiency and consistency).	Data warehouse or data mart (a non-normalized data repository primarily focused on accuracy and completeness).
Reporting	Routine, periodic, narrowly focused reports.	Ad hoc, multidimensional, broadly focused reports and queries.
Resource requirements	Ordinary relational databases.	Multiprocessor, large-capacity, specialized databases.
Execution speed	Fast (recording of business transactions and routine reports).	Slow (resource intensive, complex, large-scale queries).

OLAP OPERATIONS

Slice – a subset of a multidimensional array

Dice – a slice on more than two dimensions

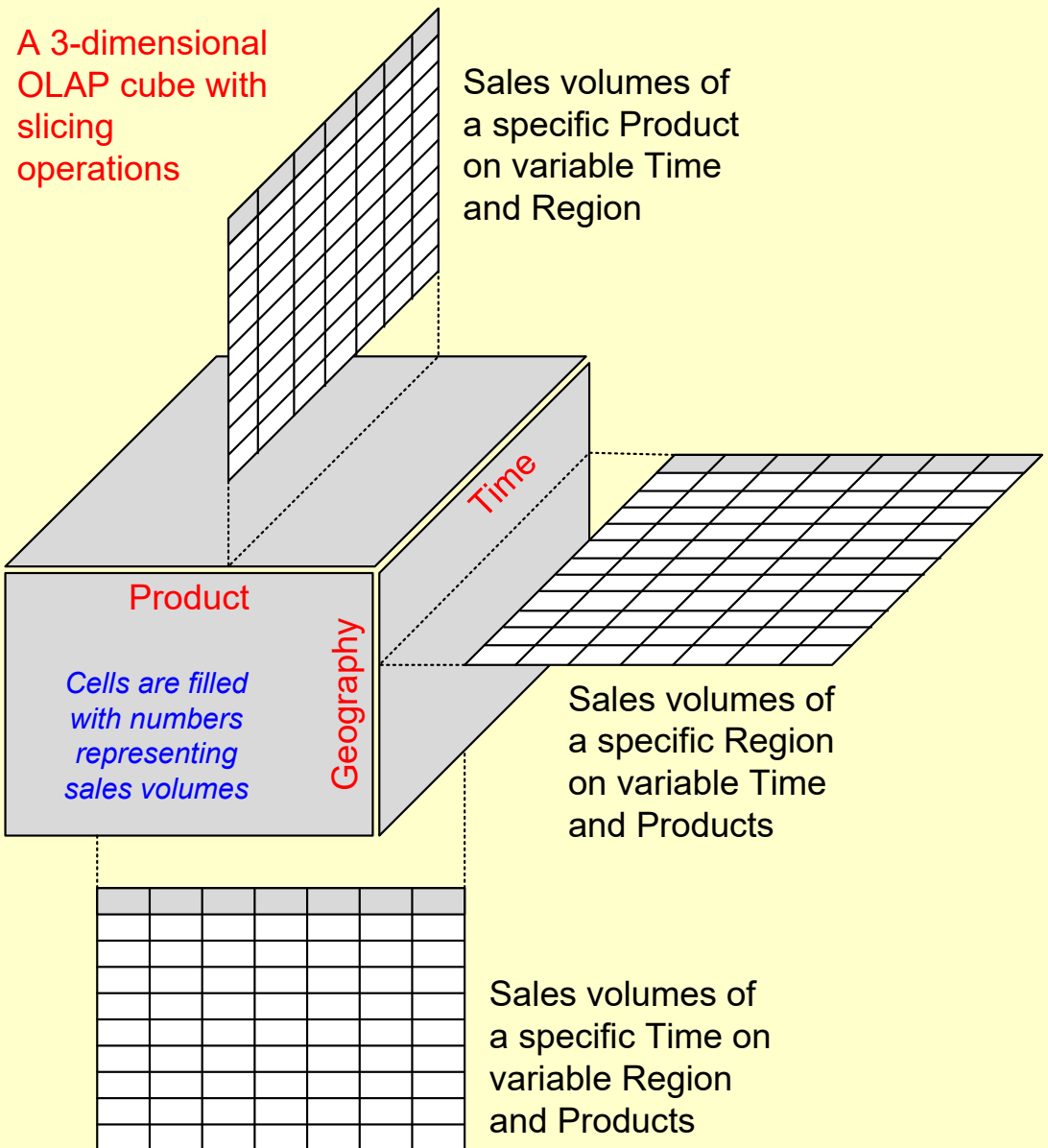
Drill Down/Up – navigating among levels of data ranging from the most summarized (up) to the most detailed (down)

Roll Up – computing all of the data relationships for one or more dimensions

Pivot – used to change the dimensional orientation of a report or an ad hoc query-page display

OLAP

Slicing Operations on a Simple Tree-Dimensional Data Cube



VARIATIONS OF OLAP

Multidimensional OLAP (MOLAP)

OLAP implemented via a specialized multidimensional database (or data store) that summarizes transactions into multidimensional views ahead of time

Relational OLAP (ROLAP)

The implementation of an OLAP database on top of an existing relational database

Database OLAP and Web OLAP (DOLAP and WOLAP); Desktop OLAP,...

DW IMPLEMENTATION ISSUES

11 tasks for successful DW implementation

- Establishment of service-level agreements and data-refresh requirements
- Identification of data sources and their governance policies
- Data quality planning
- Data model design
- ETL tool selection
- Relational database software and platform selection
- Data transport
- Data conversion
- Reconciliation process
- Purge and archive planning
- End-user support



DW IMPLEMENTATION GUIDELINES

Project must fit with corporate strategy & business objectives

There must be complete buy-in to the project by executives, managers, and users

It is important to manage user expectations about the completed project

The data warehouse must be built incrementally

Build in adaptability, flexibility and scalability

The project must be managed by both IT and business professionals

Only load data that have been cleansed and are of a quality understood by the organization

Do not overlook training requirements

Be politically aware

SUCCESSFUL DW IMPLEMENTATION

THINGS TO AVOID

Starting with the wrong sponsorship chain

Setting expectations that you cannot meet

Engaging in politically naive behavior

Loading the data warehouse with information just because it is available

Believing that data warehousing database design is the same as transactional database design

Choosing a data warehouse manager who is technology oriented rather than user oriented

SUCCESSFUL DW IMPLEMENTATION

THINGS TO AVOID - CONT.

Focusing on traditional internal record-oriented data and ignoring the value of external data and of text, images, etc.

Delivering data with confusing definitions

Believing promises of performance, capacity, and scalability

Believing that your problems are over when the data warehouse is up and running

Focusing on ad hoc data mining and periodic reporting instead of alerts

FAILURE FACTORS IN DW PROJECTS

Lack of executive sponsorship

Unclear business objectives

Cultural issues being ignored

- Change management

Unrealistic expectations

Inappropriate architecture

Low data quality / missing information

Loading data just because it is available

MASSIVE DW AND SCALABILITY

Scalability

- The main issues pertaining to scalability:
 - The amount of data in the warehouse
 - How quickly the warehouse is expected to grow
 - The number of concurrent users
 - The complexity of user queries
- Good scalability means that queries and other data-access functions will grow linearly with the size of the warehouse

REAL-TIME/ACTIVE DW/BI

Enabling real-time data updates for real-time analysis and real-time decision making is growing rapidly

- Push vs. Pull (of data)

Concerns about real-time BI

- Not all data should be updated continuously
- Mismatch of reports generated minutes apart
- May be cost prohibitive
- May also be infeasible

REAL-TIME/ACTIVE DW AT TERADATA

Active Access

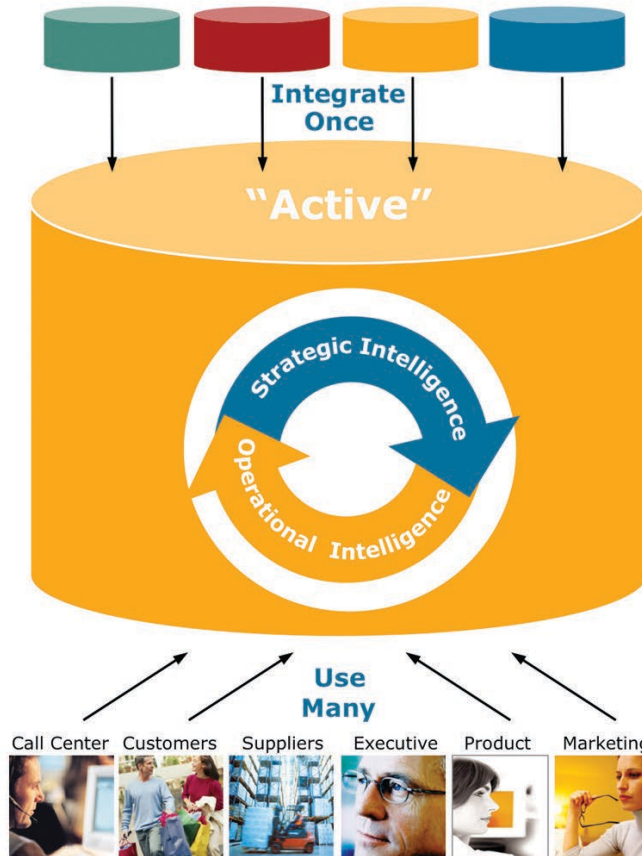
Front-Line operational decisions or services supported by near-real-time (NRT) access; Service Level Agreements of 5 seconds or less

Active Load

Intra-day data acquisition; Mini-batch to NRT trickle data feeds measured in minutes or seconds

Active Events

Proactive monitoring of business activity initiating intelligent actions based on rules and context; to systems or users supporting an operational business process



Active Workload Management

Dynamically manage system resources for optimum performance and resource utilization supporting a mixed-workload environment

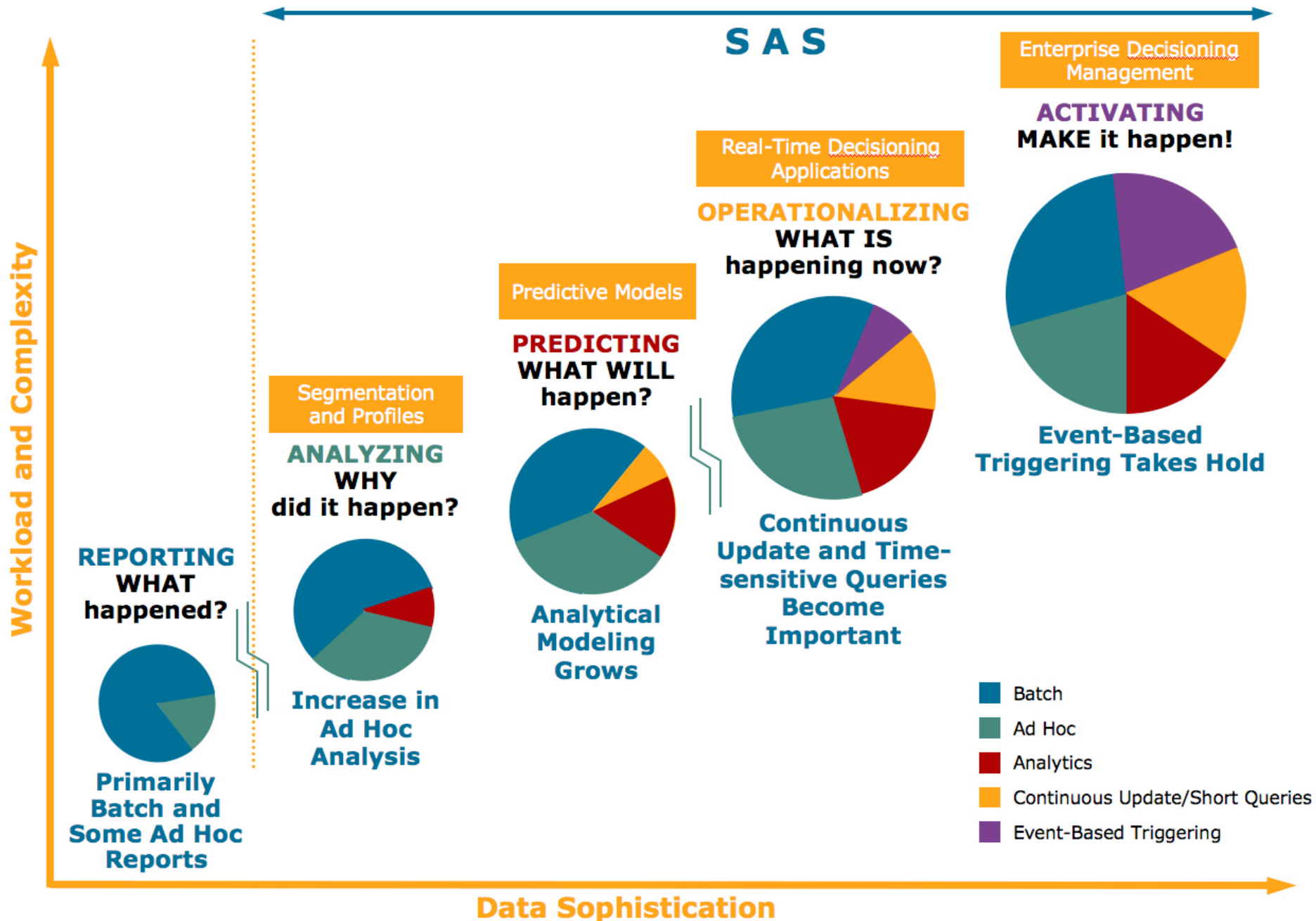
Active Enterprise Integration

Integration into the Enterprise Architecture for delivery of intelligent decisioning services

Active Availability

Business Continuity to support the requirements of the business (up to 7X24)

ENTERPRISE DECISION EVOLUTION AND DW



TRADITIONAL VS ACTIVE DW ENVIRONMENT

Traditional Data Warehousing Environment

Strategic decisions only
Results sometimes hard to measure
Daily, weekly, monthly data currency is acceptable; summaries are often appropriate
Moderate user concurrency

Highly restrictive reporting used to confirm or check existing processes and patterns; often uses predeveloped summary tables or data marts

Power users, knowledge workers, internal users

Active Data Warehousing Environment

Strategic and tactical decisions
Results measured with operations
Only comprehensive detailed data available within minutes is acceptable
High number (1,000 or more) of users accessing and querying the system simultaneously

Flexible ad hoc reporting as well as machine-assisted modeling (e.g., data mining) to discover new hypotheses and relationships

Operational staffs, call centers, external users

DW ADMINISTRATION AND SECURITY

Data warehouse administrator (DWA)

- DWA should...
 - have the knowledge of high-performance software, hardware and networking technologies.
 - possess solid business knowledge and insight.
 - be familiar with the decision-making processes so as to suitably design/maintain the data warehouse structure.
 - possess excellent communications skills.

Security and privacy is a pressing issue in DW

- Safeguarding the most valuable assets
- Government regulations (HIPAA, etc.)
- Must be explicitly planned and executed

THE FUTURE OF DW

Sourcing...

- Open source software
- SaaS (software as a service)
- Cloud computing
- DW appliances

Infrastructure...

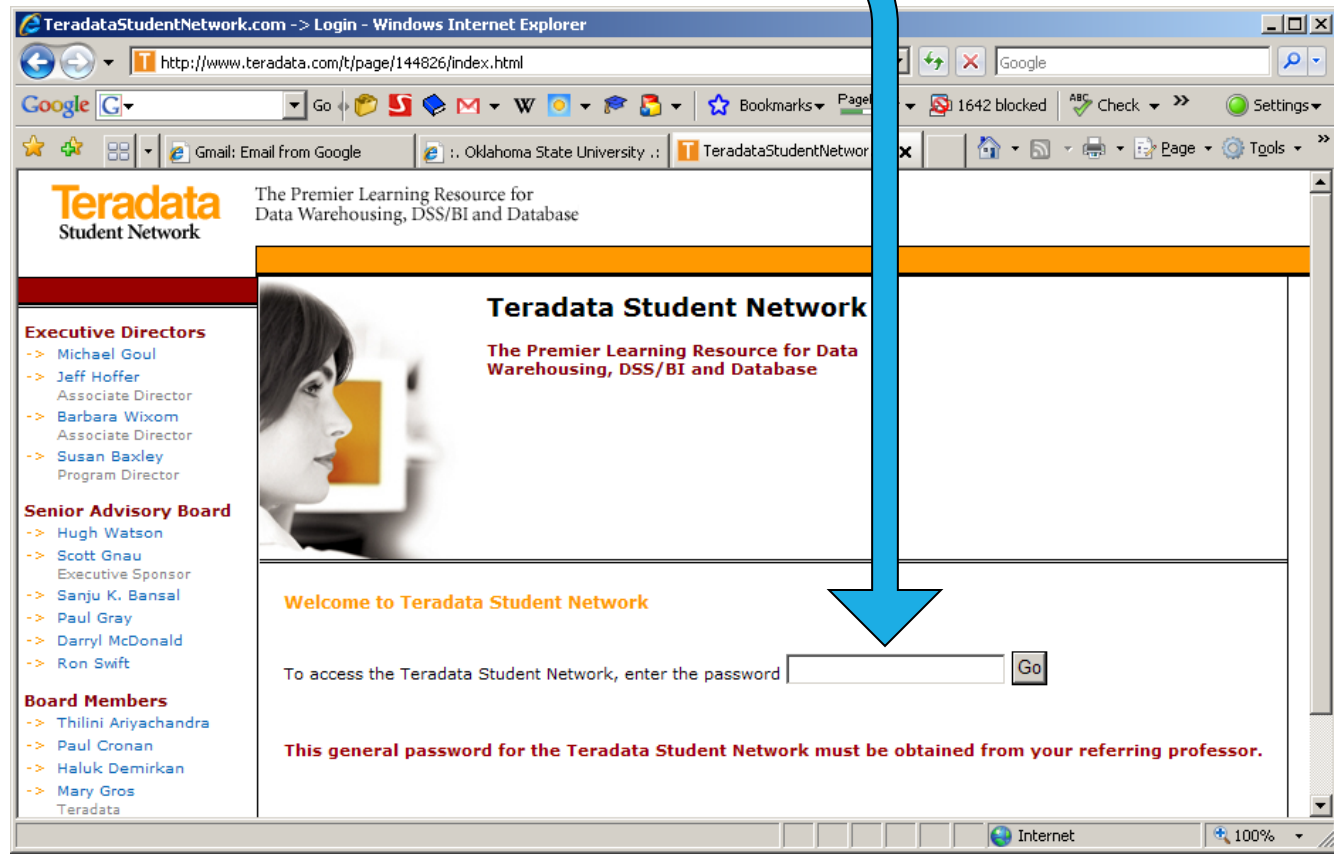
- Real-time DW
- Data management practices/technologies
- In-memory processing (“super-computing”)
- New DBMS
- Advanced analytics

BI / OLAP PORTAL FOR LEARNING

MicroStrategy, and much more...

www.TeradataStudentNetwork.com

Pw: <check with TDUN>





END OF THE CHAPTER

Questions, comments